Table of Contents

CHAPTER 1: ABSTRACT **3**

CHAPTER 2: INTRODUCTION **4**

2.1 Overview…….. **4**

2.2 Problem Statement ..... **Error! Bookmark not defined.**

2.3 Reasearch Motivation…… 5

2.4 Research Objectives…… 6

2.5 Signficance…… 7

2.6 Advantages…… 7

2.7 Applications ..... **Error! Bookmark not defined.**

CHAPTER 3: LITERATURE SURVEY **9**

3.1 Survey of Relaterd Literature 9

3.2 Review of Related Work ..... **Error! Bookmark not defined.**

3.3 Research Gap ..... **Error! Bookmark not defined.**

CHAPTER 4: PROBLEM STATEMENT **12**

4.1 Statement of the Problem 12

4.2 Key Challenges ..... **Error! Bookmark not defined.**

4.3 Need for the Proposed Solution ..... **Error! Bookmark not defined.**

5.1 Description of the Existing System 14

5.2 Tools Used Today ..... **Error! Bookmark not defined.**

5.3 Disadvantages ..... **Error! Bookmark not defined.**

CHAPTER 6: PROPOSED SYSTEM **16**

6.1 Overview 16

6.2 Key Features 16

6.3 Working Principle ..... **Error! Bookmark not defined.**

6.4 Advantages ..... **Error! Bookmark not defined.**

6.5 Comparison with the Existing System ..... **Error! Bookmark not defined.**

6.6 Feasibility Study ..... **Error! Bookmark not defined.**

7. HARDWARE REQUIREMENTS **19**

7.1 Hardware Specification 19

7.2 Remarks 19

8. SOFTWARE REQUIREMENTS **19**

8.1 Software Specification 20
8.2 Functional Requirements 20
8.3 Non-Functional Requirements 21
9. SYSTEM ARCHITECTURE 22
9.1 Design Goals 22
9.2 System Architecture ..... **Error! Bookmark not defined.**
9.3 Layered View ..... **Error! Bookmark not defined.**
9.4 Data Flow 23
9.5 Document Processing and Retrieval Design ..... **Error! Bookmark not defined.**
9.6 Grounded Generation and Hallucination Control ..... **Error! Bookmark not defined.**
9.7 Database Design ..... **Error! Bookmark not defined.**
9.8 Security and Privacy ..... **Error! Bookmark not defined.**
10. MODULES 25
10.1 Module Overview 25
10.2 Description of Modules 26
Module 1: User Management 26
Module 2: Document Processing ..... **Error! Bookmark not defined.**
Module 3: Semantic Search and Vector Database ..... **Error! Bookmark not defined.**
Module 4: AI Analysis and RAG ..... **Error! Bookmark not defined.**
Module 5: Research Analysis ..... **Error! Bookmark not defined.**
Module 6: Literature Review ..... **Error! Bookmark not defined.**
Module 7: Visualization and Content Generation ..... **Error! Bookmark not defined.**
11. UML DIAGRAMS 28
11.1 Use Case Diagram 28
11.2 Class Diagram 29
11.3 Sequence Diagram 30
REFERENCES 32

CHAPTER 1: ABSTRACT

The AI-Powered Career Roadmap and Skill Recommendation System is a smart career guidance application developed to assist students and job seekers in selecting suitable career paths and identifying the skills required to achieve their professional goals. Many students face difficulty in choosing the right career because of limited awareness about available career opportunities, changing industry requirements, and uncertainty about the skills they need to develop. The proposed system uses Artificial Intelligence and Machine Learning techniques to analyze user details such as educational background, skills, interests, and career preferences. Based on this information, the system recommends suitable career options and provides a personalized roadmap containing the important technical and soft skills required for the selected career. It also helps users identify their existing skill gaps and provides guidance on areas that need improvement. The system aims to reduce confusion in career selection, save time in searching for relevant career information, and provide a structured approach to skill development. With a simple and user-friendly interface, the proposed system provides personalized career recommendations and supports users in planning their learning and career journey effectively. Overall, the system serves as an intelligent career guidance platform that helps students and job seekers make informed career decisions and improve their employability in a competitive job market.

Keywords: Artificial Intelligence, Machine Learning, Career Guidance, Career Recommendation, Skill Recommendation, Career Roadmap, Skill Gap Analysis, Personalized Learning, Employability.

CHAPTER 2: INTRODUCTION

2.1 Overview

Career selection is one of the most important decisions in a student's life. Students and graduates often face difficulties in identifying a suitable career path because of limited awareness about available job roles, changing industry requirements, and uncertainty about the skills needed for their desired careers. Although there are many sources of career-related information available today, finding the right information and connecting it with an individual's education, skills, interests, and career goals can be challenging. Traditional career guidance methods may provide general suggestions, but they may not always offer personalized guidance according to the specific needs of each individual.

The rapid growth of technology and Artificial Intelligence has created opportunities to improve the way career guidance is provided. AI-based systems can analyze user-related information and provide recommendations according to the user's profile. In the field of career planning, such systems can help individuals explore suitable career options, understand the skills required for different roles, and identify areas where they need further improvement. This can make the career decision-making process more organized and easier to understand.

The **AI-Powered Career Roadmap and Skill Recommendation System** is proposed to address these challenges by providing personalized career guidance. The system considers relevant information about the user, such as educational background, existing skills, interests, and career preferences, to support career recommendations. Along with suggesting possible career paths, the system focuses on identifying the skills that are important for those career paths. This helps users understand the relationship between their current abilities and the requirements of their desired career.

Another important aspect of the proposed system is the generation of a career roadmap. Simply knowing which career to choose may not be sufficient for a student. Users also need to understand what they should learn and how they can gradually develop the required skills. A structured roadmap can provide a clear direction for skill development and help users focus on relevant areas. It can also help reduce confusion caused by the large amount of career and learning information available online.

The system is particularly useful for students and fresh graduates who are beginning their career journey. It can help them make better-informed decisions by presenting career and skill-related information in a personalized and organized manner. Instead of following the same learning path as everyone else, users can focus on skills that are relevant to their interests and career objectives. The system can therefore support continuous learning and career development.

The proposed system aims to combine **career recommendation, skill recommendation, skill-gap awareness, and career roadmap generation** into a single platform. By using Artificial Intelligence to support the recommendation process, the system provides a systematic approach to career planning. It is intended to act as a decision-support system that assists users in understanding possible career opportunities and planning the skills they need to develop. Overall, the project aims

to make career planning more personalized, structured, and accessible for students, graduates, and other individuals who are seeking appropriate career directions.

2.2 Problem Statement

The major problems addressed by the proposed **AI-Powered Career Roadmap and Skill Recommendation System** are:

1.Lack of Personalized Career Guidance

Students often receive general career advice that may not consider their individual educational background, skills, interests, and career preferences. This makes it difficult to identify a career path that is suitable for them.

2.Difficulty in Choosing the Right Career

There are numerous career options available in different fields. Students may not have sufficient knowledge about different job roles, their requirements, and the career opportunities associated with them, leading to confusion while making career decisions.

3.Identification of Skill Gaps

Users may not know the difference between their existing skills and the skills required for their desired career. Identifying these skill gaps is important for planning effective skill development.

4.Lack of a Structured Career Roadmap

Even after selecting a career, students may not know what skills they should learn first or what steps they should follow. The absence of a structured roadmap can make the learning and career preparation process difficult.

5.Changing Industry Requirements

Industry requirements and job roles continue to change with advancements in technology. Students need to continuously develop relevant skills to remain prepared for current and future career opportunities.

6.Information Overload

A large amount of career and learning information is available through different sources. Finding relevant information and organizing it according to an individual's career goal can be time-consuming and confusing.

Therefore, the proposed system aims to address these problems by providing personalized career recommendations, relevant skill recommendations, skill-gap awareness, and a structured career roadmap based on the user's profile.

2.3 Research Motivation

The motivation for developing the **AI-Powered Career Roadmap and Skill Recommendation System** comes from the difficulties faced by students and graduates while planning their careers. Many students complete their education without having a clear understanding of which career path is suitable for them or what skills they need to develop to achieve their career goals.

- **Need for Personalized Career Guidance:** Different students have different educational backgrounds, skills, interests, and career goals. Therefore, personalized guidance can help users identify career options that are more relevant to their individual profiles.
- **Bridging the Skill Gap:** There can be a difference between the skills a student currently possesses and the skills required for a particular career. Identifying this gap can help students focus on the areas that require improvement.
- **Providing Clear Career Direction:** Students may feel confused when there are many career options available. A systematic recommendation approach can help them understand possible career paths and make more informed decisions.
- **Guiding Skill Development:** Knowing a career goal alone is not sufficient. Students also need to know which skills they should develop. The proposed system is motivated by the need to provide relevant skill recommendations according to the selected career path.
- **Creating a Structured Roadmap:** Students often lack a clear step-by-step plan for reaching their career goals. A career roadmap can provide a structured direction for learning and skill development.
- **Use of Artificial Intelligence:** AI provides an opportunity to analyze user-related information and support personalized recommendations. Applying AI to career guidance can make the process more systematic and user-oriented.

2.4 Research Objectives

The main objectives of the **AI-Powered Career Roadmap and Skill Recommendation System** are:

- To develop an AI-powered career guidance system that assists users in planning their career paths.
- To analyze user profiles based on relevant information such as educational background, existing skills, interests, and career preferences.
- To recommend suitable career paths according to the user's profile and career interests.
- To identify required skills associated with the recommended career paths.
- To identify skill gaps by helping users understand the difference between their existing skills and the skills required for their desired career.
- To provide personalized skill recommendations that can help users focus on relevant areas of skill development.
- To generate a structured career roadmap that provides users with a clear direction for progressing toward their career goals.
- To support better career decision-making by presenting career and skill-related information in an organized and understandable manner.

- To assist students and graduates in planning their career development and preparing for future employment opportunities.
- To provide a flexible system that can be further enhanced with additional career roles, skills, and recommendation capabilities in the future.

2.5 Significance

The significance of this project lies in providing a personalized and systematic approach to career planning. Students and graduates often have difficulty understanding which career path is suitable for them and what skills they need to develop. The proposed system addresses this need by bringing career recommendations, skill recommendations, and career roadmap generation together in a single platform.The system can help users make more informed career decisions by considering their individual profile, interests, existing skills, and career preferences. It also helps users understand the skills associated with their desired career and recognize areas where additional skill development may be required. This can provide better awareness of the user's current position and the requirements of the selected career.

2.6 Advantages

The proposed AI-Powered Career Roadmap and Skill Recommendation System **provides several advantages for students, graduates, and other users seeking career guidance.**

- **Personalized Career Recommendations:** Provides career suggestions based on the user's profile, interests, skills, and career preferences.
- **Skill Recommendations:** Helps users understand the skills that are relevant to their selected or recommended career path.
- **Skill Gap Awareness:** Helps users identify areas where their existing skills may need further development.
- **Structured Career Roadmap:** Provides a clear direction for developing relevant skills and progressing toward a desired career.
- **Reduced Career Confusion:** Organizes career and skill-related information in a simple manner, helping users make better-informed decisions.
- **Time Saving:** Reduces the need to search through multiple sources for basic career and skill information.
- **Easy Accessibility:** Provides career-related guidance through a single system, making the information easier to access and understand.
- **Supports Career Development:** Helps users plan their learning and skill development according to their career goals.
- **Future Expandability:** The system can be extended in the future by adding more career roles, skills, and recommendation capabilities.

2.7 Applications

he AI-Powered Career Roadmap and Skill Recommendation System can be applied in various areas related to education, career development, and skill enhancement. The system helps users understand the skills required for their desired career and provides suitable guidance for improving their knowledge and employability.

The major applications of the proposed system include:

1. **Career Planning**: Helps students and job seekers identify suitable career paths based on their interests, skills, and career goals.
2. **Skill Gap Identification:** Helps users understand the difference between their existing skills and the skills required for a particular career or job role.
3. **Personalized Learning:** Recommends relevant skills and learning areas so that users can focus on the knowledge required for their selected career path.
4. **Student Career Guidance**: Can be used by educational institutions to provide career guidance and support students in making informed career decisions.
5. **Job Preparation**: Helps users identify the technical and professional skills they need to develop before applying for specific job roles.
6. **Career Roadmap Development**: Provides a structured direction for users to progress from their current skill level toward their desired career.
7. **Skill Development Programs:** Training institutes and organizations can use the system to identify common skill requirements and guide learners toward suitable development programs.

   Overall, the system can serve as a useful career-support tool by providing personalized recommendations and a structured roadmap for continuous skill

CHAPTER 3: LITERATURE SURVEY

3.1 Survey of Related Literature

Career selection and skill development have become important challenges for students and job seekers due to the continuous changes in technology and employment requirements. Traditional career guidance mainly depends on counselling, aptitude tests, questionnaires, and general information about different professions. Although these approaches provide basic guidance, they may not provide personalized recommendations according to the individual skills, interests, qualifications, and career goals of a user.

Several research studies have explored the use of Artificial Intelligence, Machine Learning, recommendation systems, and data analysis for improving career guidance. AI-based career recommendation systems attempt to analyse information about users and identify career options that are more suitable for their profiles. These systems can consider factors such as educational background, interests, skills, academic performance, and career preferences.

Research on **career recommendation systems** has shown that recommendation techniques can help users select suitable career options from a large number of available choices. Instead of depending completely on manual counselling, automated systems can process user information and provide recommendations in a shorter period of time. Such systems can be particularly useful for students who are uncertain about their career direction.

Another important area of research is **skill recommendation**. Different career roles require different combinations of technical and professional skills. Skill recommendation systems attempt to identify the skills that are relevant to a particular career or job role. These systems can help users understand which skills they already possess and which additional skills they need to develop.

Studies related to **skill-gap analysis** focus on comparing an individual's existing skills with the skills required for a particular job role. Identifying this gap can help users plan their learning activities more effectively. Instead of learning skills randomly, users can concentrate on the areas that are most relevant to their career objectives.

Research in **personalized learning and recommendation systems** has also demonstrated the importance of providing recommendations according to individual user requirements. A personalized system can provide different recommendations to different users instead of giving the same information to everyone. This approach can make career and learning guidance more relevant and useful.

Another related concept is the development of a **career roadmap**. A career roadmap provides a structured sequence of skills and learning areas that a user can follow to progress toward a desired career. It provides direction regarding what should be learned first and what skills can be developed at later stages. Combining career recommendations with a roadmap can therefore provide more complete guidance than simply suggesting a career.

The literature related to AI-based recommendation, skill analysis, personalized learning, and career planning provides the foundation for the development of the proposed system. These research areas demonstrate that intelligent recommendation techniques can be used to support users in making better career decisions and planning their skill development.

3.2 Detailed Analytical Review of Literature

The existing literature indicates that technology-based career guidance systems have gradually developed from simple information-providing systems into intelligent recommendation systems. Traditional systems mainly provide information about different careers, qualifications, and job opportunities. However, they generally do not provide sufficient personalization for individual users.

AI-based career recommendation approaches provide an improvement by considering multiple user-related factors before generating career recommendations. The use of intelligent techniques allows systems to identify relationships between user characteristics and career requirements. This can help users receive recommendations that are more closely related to their individual profiles.

Skill recommendation systems provide another important improvement. A career recommendation alone may not be sufficient for a student or job seeker because the user may not know how to prepare for the recommended career. Skill recommendation addresses this issue by identifying the knowledge and abilities required for a particular career.

Skill-gap analysis further improves the recommendation process by comparing the user's current skills with the expected skills for a target career. This comparison can help determine which areas require additional development. For example, if a user has basic knowledge of a particular technical area but lacks advanced skills required for a job role, the system can highlight the areas that need improvement.

Personalization is another important aspect identified from the literature. Users have different educational backgrounds, interests, existing skills, and career goals. Therefore, providing identical recommendations to every user may not be effective. A personalized recommendation approach can make the guidance more relevant by considering the information provided by each individual.

The literature also indicates that career planning should not end with recommending a career. Users need a clear direction regarding how they can reach their selected career. A roadmap can provide this direction by organizing required skills into a logical progression. This makes the learning process easier to understand and can help users track their progress.

From the overall analysis, it can be observed that career recommendation, skill recommendation, skill-gap analysis, and roadmap generation are closely related but are often handled as separate functions. Integrating these functions into one system can provide a more comprehensive solution for career planning.

The proposed system is therefore designed around the idea of connecting these important components. It aims to take user information as input, provide suitable career recommendations, identify relevant skills, and generate a structured career roadmap. This integrated approach can provide users with a clearer understanding of both **where they want to go** and **what they need to learn to get there**.

3.3 Research Gap Analysis

The review of existing literature identifies several gaps in current career guidance and recommendation approaches. Many existing systems concentrate primarily on recommending

suitable careers, while limited attention may be given to the complete process of preparing the user for the selected career.

One major research gap is the **lack of integration between career recommendation and skill recommendation**. A system may suggest a career but may not clearly explain the skills required to achieve that career. This can leave users uncertain about their next steps.

Another gap is the limited integration of **skill-gap identification** with career recommendations. Knowing the required skills is useful, but users also need to understand which of those skills they currently lack. A comparison between existing and required skills can provide more meaningful guidance.

A further gap is the absence of a **structured and personalized career roadmap** in some existing approaches. Users may receive recommendations without receiving a clear sequence for developing the necessary skills. A roadmap can help overcome this issue by providing a systematic direction for skill development.

Existing approaches may also depend on static information or predefined recommendations. Since career requirements and technology trends continue to change, a career guidance system should be designed to provide relevant recommendations based on the available career and skill information.

The proposed **AI-Powered Career Roadmap and Skill Recommendation System** attempts to address these gaps by bringing multiple career guidance functions together. The system focuses on:

- Providing career recommendations based on user-related information.
- Identifying skills relevant to the recommended career.
- Helping users understand the skills they need to develop.
- Providing a structured roadmap for career preparation.
- Offering personalized guidance instead of only providing general career information.
- Connecting career selection with skill development in a single system.

Thus, the identified research gaps provide the motivation for developing the proposed system. The system aims to provide a more organized and user-oriented approach to career planning by integrating **career recommendation, skill recommendation, and career roadmap generation** into one platform.

This section reviews the research on which ResearchMate AI is built. The papers are grouped by the part of the system they support: retrieval-augmented generation, language understanding and long documents, summarization, semantic search, and the reliability of generated text.

CHAPTER 4: PROBLEM STATEMENT

4.1 Statement of the Problem

Choosing the right career is a major challenge for students and job seekers because there are many career options available, each requiring different educational qualifications, technical skills, and professional abilities. Many students are unsure about which career is suitable for their interests, education, and existing skills. As a result, they may choose a career without having a clear understanding of its requirements.

Existing career guidance systems mainly provide general career information or recommendations. However, they may not effectively connect the recommended career with the specific skills that a user needs to develop. Users often have to search through multiple websites and learning platforms to identify the required skills and decide what they should learn first.

Another major difficulty is the absence of a structured career roadmap. Even after identifying a suitable career, users may not know the proper sequence in which they should develop their skills. This can lead to confusion, inefficient learning, and difficulty in planning their career development.

Therefore, there is a need for an **AI-powered integrated system** that can analyse user information, recommend suitable career options, identify the skills required for those careers, highlight potential skill gaps, and generate a structured career roadmap. The proposed **AI-Powered Career Roadmap and Skill Recommendation System** aims to address these problems by providing personalized and systematic career guidance through a single platform.

The major problems addressed by the proposed **AI-Powered Career Roadmap and Skill Recommendation System** are:

1.Lack of Personalized Career Guidance

Students often receive general career advice that may not consider their individual educational background, skills, interests, and career preferences. This makes it difficult to identify a career path that is suitable for them.

2.Difficulty in Choosing the Right Career

There are numerous career options available in different fields. Students may not have sufficient knowledge about different job roles, their requirements, and the career opportunities associated with them, leading to confusion while making career decisions.

3.Identification of Skill Gaps

Users may not know the difference between their existing skills and the skills required for their desired career. Identifying these skill gaps is important for planning effective skill development.

4.Lack of a Structured Career Roadmap

Even after selecting a career, students may not know what skills they should learn first or what steps they should follow. The absence of a structured roadmap can make the learning and career preparation process difficult.

5.Changing Industry Requirements

Industry requirements and job roles continue to change with advancements in technology. Students need to continuously develop relevant skills to remain prepared for current and future career opportunities.

6.Information Overload

A large amount of career and learning information is available through different sources. Finding relevant information and organizing it according to an individual's career goal can be time-consuming and confusing.

Therefore, the proposed system aims to address these problems by providing personalized career recommendations, relevant skill recommendations, skill-gap awareness, and a structured career roadmap based on the user's profile.

CHAPTER 5: EXISTING SYSTEM

5.1 Existing Career Guidance Systems

Career guidance systems are designed to help students, graduates, and job seekers make decisions about their education, skills, and career choices. Traditionally, career guidance is provided through teachers, career counsellors, educational institutions, aptitude tests, and online career portals. These methods provide users with information about different career options, educational qualifications, job roles, and required skills.

Many existing digital career guidance systems use questionnaires or user-provided information to suggest suitable career options. Users may enter details such as their educational background, interests, strengths, skills, and preferred career areas. The system then compares the provided information with predefined career profiles and provides recommendations.

Some online platforms focus mainly on **career information**, where users can search for different job roles and view information about qualifications, responsibilities, and required skills. Other systems use recommendation techniques to suggest careers based on user interests and abilities.

There are also systems that focus specifically on **skill development**. These platforms recommend courses, learning materials, or individual skills based on a selected career or job role. Such systems can help users improve their employability by identifying areas that are relevant to their desired profession.

Existing career guidance approaches can therefore be broadly viewed as systems that provide:

- Career information and job-role descriptions.
- Career recommendations based on user profiles.
- Skill and course recommendations.
- Educational and professional guidance.
- Information about qualifications and career requirements.
- Basic assessment of user interests and abilities.

These systems have made career information more accessible and have reduced the dependence on traditional counselling methods. However, many existing approaches provide only a particular part of the overall career planning process. A user may need to use different platforms for career selection, skill identification, learning resources, and career planning.

The proposed system aims to provide a more integrated approach by connecting career recommendation, skill recommendation, and career roadmap generation within a single system.

Automated and semi-automated decision-making in administrative, financial, and educational domains has historically relied on three main paradigms: manual human review, rule-based expert systems, and single-model machine learning classifiers.

5.2 Manual Decision-Making

Although existing career guidance systems provide useful information and recommendations, they have several limitations. These limitations can make it difficult for users to develop a complete and personalized career plan.

- **Limited personalization:** Some systems provide similar recommendations to users with similar interests without sufficiently considering their individual skill levels and career goals.
- **Career recommendation without a roadmap:** Many systems may suggest suitable career options but do not provide a clear step-by-step roadmap for achieving the selected career.
- **Separate skill recommendations:** Skill-related recommendations may be available separately from career recommendations, requiring users to search for the required skills on different platforms.
- **Limited skill-gap identification:** Existing systems may identify required skills but may not clearly compare them with the user's existing skills to determine the areas that need improvement.
- **Dependence on predefined information:** Some systems rely heavily on predefined career profiles and databases, which may limit the flexibility of recommendations.
- **Lack of integrated guidance:** Career selection, skill development, and roadmap planning are often treated as separate activities rather than as connected stages of the same process.
- **Difficulty for beginners:** Students who have little knowledge about different career options may find it difficult to understand what they should learn first and how they should progress toward a career.

These limitations show the need for an integrated career guidance solution that can provide users with career recommendations along with relevant skills and a structured roadmap. The proposed **AI-Powered Career Roadmap and Skill Recommendation System** is intended to address these limitations by providing a more organized and personalized approach to career

CHAPTER 6: PROPOSED SYSTEM

6.1 Overview

The proposed system is an **AI-Powered Career Roadmap and Skill Recommendation System** designed to assist students and job seekers in making informed career decisions. The system provides personalized career guidance by considering the user's information, interests, existing skills, and career preferences.

The main purpose of the proposed system is to overcome the limitations of existing career guidance systems by integrating career recommendation, skill recommendation, and career roadmap generation into a single platform. Instead of providing only a list of career options, the system helps users understand the skills associated with their selected career and provides a structured direction for developing those skills.

The system accepts relevant user information as input and processes it to identify suitable career options. After a career is recommended or selected, the system identifies the important skills associated with that career. Based on these requirements, a roadmap can be generated to help the user understand the sequence of skills and learning areas that should be developed.

The proposed system can be useful for students who are uncertain about their career direction, graduates preparing for employment, and individuals who want to improve their skills for a particular career.

6.2 System Modules

The proposed system is divided into different modules to perform specific functions. Each module contributes to the overall career recommendation and roadmap generation process.

User Input Module

The User Input Module collects the information required from the user. This may include educational background, interests, existing skills, career preferences, and other relevant information.

The collected information forms the basis for generating personalized recommendations.

User Profile Module

The User Profile Module organizes the information provided by the user into a structured profile. This profile represents the user's current skills, interests, educational background, and career preferences.

The profile is used by the recommendation components to generate relevant results.

Career Recommendation Module

The Career Recommendation Module analyses the user profile and identifies career options that are relevant to the user's information and preferences.

The purpose of this module is to help users discover career paths that match their profile rather than selecting a career without sufficient information.

Skill Recommendation Module

The Skill Recommendation Module identifies skills that are important for the recommended or selected career. It helps the user understand which technical and professional skills should be developed.

This module connects the selected career with its required skills.

Career Roadmap Module

The Career Roadmap Module generates a structured roadmap for the user. It organizes the recommended skills and career-related learning areas into a logical progression.

The roadmap provides users with a clearer direction for developing their skills and preparing for their desired career.

Results Module

The Results Module presents the final recommendations to the user. It displays information such as the recommended career, relevant skills, and generated career roadmap in an understandable format

6.3 Career Recommendation Process

The career recommendation process is one of the major functions of the proposed system. It is designed to help users identify career options that are relevant to their profile.The process begins when the user provides the required information through the system interface. The system processes the information and compares the user's profile with available career-related information.Relevant factors are considered during the recommendation process, such as the user's interests, existing skills, educational background, and career preferences. Based on the available information, the system identifies suitable career options.The recommended careers are then presented to the user. The user can review the recommendations and select a career that matches their interests and future goals.

Career recommendation flow:

User Input → Profile Analysis → Career Matching → Career Recommendation → User Selection

This process helps users make more informed career decisions instead of depending only on assumptions or general career information.

6.4 Skill Recommendation Process

After identifying a suitable career, the system focuses on the skills required for that career. Different career roles require different combinations of technical and professional skills. Therefore, understanding the required skills is an important part of career preparation.The Skill Recommendation Process takes the selected career as an important input and identifies the skills

associated with it. The system then presents the relevant skills to the user.The recommended skills help users understand what they need to learn or improve to prepare for their desired career. This can also help users identify areas where their existing knowledge may be insufficient.

Skill recommendation flow:

Selected Career → Required Skill Identification → Skill Matching → Skill Recommendations → Skill Development

The process provides users with a clear connection between their career goal and the skills they need to develop.

6.5 Career Roadmap Generation

The Career Roadmap Generation module provides a structured direction for users to progress toward their selected career. Simply knowing the required skills may not be sufficient because users may not know the order in which they should develop those skills.The system organizes relevant skills and learning areas into a logical sequence. The roadmap begins with foundational areas and progresses toward more advanced or career-specific skills, depending on the selected career.The generated roadmap helps users understand what they should focus on during their career preparation. It can also provide a structured approach to continuous skill development.

Roadmap generation flow:

Selected Career → Required Skills → Skill Prioritization → Learning Progression → Career Roadmap

The roadmap acts as a guide for users and provides a clearer path from their current learning stage toward their desired career.

6.6 Advantages and Limitations

Advantages

- Provides personalized career recommendations.
- Connects career selection with relevant skill recommendations.
- Helps users understand the skills required for their desired career

Limitations:

- The quality of recommendations depends on the information available to the system.
- Recommendations may not always perfectly represent an individual's personal circumstances.
- Career requirements can change over time as industry trends evolve.
- The system provides guidance and recommendations but cannot completely replace professional career counselling.

CHAPTER 7: HARDWARE REQUIREMENTS

7.1 Hardware Specification

The **AI-Powered Career Roadmap and Skill Recommendation System** can be developed and demonstrated using a standard computer or laptop. The hardware requirements mainly depend on the processing requirements of the application and the way AI-based recommendation components are executed. Table 7.1 presents the minimum and recommended hardware configurations required for developing and running the system.

Component	Minimum	Recommended
Processor	Quad-core CPU (Intel i5 / Ryzen 5 class)	Octa-core CPU (Intel i7 / Ryzen 7 class)
Memory (RAM)	8 GB	16 GB or more
Storage	20 GB free space	50 GB SSD
GPU	Not required (CPU embeddings or API-based LLM)	Optional GPU with 8 GB or more VRAM for local embedding or a local LLM
Display	1366 × 768	1920 × 1080
Network	Broadband internet connection	Stable high-speed connection for API access

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Table 7.1: Hardware requirements

7.2 Remarks

- The system can be developed and executed on a standard laptop or desktop computer without requiring dedicated high-end hardware.
- A higher amount of RAM and a faster processor can improve the overall performance of the application, especially when multiple components are running simultaneously.
- A dedicated GPU is not essential for basic development and execution of the system. It may be useful if computationally intensive AI models are executed locally.
- Adequate storage should be available for the project files, datasets, application files, and other resources used during development

CHAPTER 8: SOFTWARE REQUIREMENTS

8.1 Software Specification

The **AI-Powered Career Roadmap and Skill Recommendation System** requires a suitable software environment for development, execution, and testing. The software requirements include the operating system, programming environment, development tools, database, web technologies, and other libraries or frameworks used by the system. Table 8.1 presents the software requirements for developing and running the proposed system.

Table 8.1: Software requirements

Item	Specification
Operating system	Windows 10/11, Linux or macOS
Programming languages	Python
Front end	As implemented in the project
Back end	Python-based application
PDF processing	PyMuPDF, pdfplumber
Development Environment	Visual Studio Code / Python IDE
Internet	Required for accessing online resources and services
AI/ML Libraries	Libraries used for the recommendation system
Database	As used in the project implementation
Web Browser	Google Chrome / Microsoft Edge / equivalent
Tools	Git, VS Code, Postman, a modern web browser

8.2 Software Environment

The system is developed using a suitable programming environment that supports the implementation of the career recommendation and skill recommendation functionalities. Python can be used for implementing the application logic and AI/ML-related components. A suitable

development environment is used to write, execute, debug, and maintain the project code.

The system may also require a database for storing relevant user information and other application data, depending on the implemented architecture. A modern web browser can be used to access and interact with the application when the system is implemented as a web-based application.

8.3 Remarks

- The required software depends on the technologies and libraries used in the actual implementation of the system.
- The development environment should support the programming language and libraries required by the project.
- A modern web browser is required if the system provides a web-based user interface.
- Internet connectivity may be required for accessing external resources or online services used by the application.
- The software environment can be extended in the future if additional AI models, databases, or application features are incorporated.

CHAPTER 9: SYSTEM ARCHITECTURE

9.1 System Architecture

The **AI-Powered Career Roadmap and Skill Recommendation System** follows a structured architecture in which the user's information is processed through different stages to generate personalized career and skill recommendations. The system takes relevant user inputs such as educational background, existing skills, interests, and career preferences. These inputs are processed by the application to determine suitable career options and the skills associated with them.

The architecture consists of an **Input Layer, Processing Layer, Recommendation Layer, and Output Layer**. The Input Layer collects the information provided by the user. The Processing Layer analyzes and prepares the user information for further processing. The Recommendation Layer performs the career and skill recommendation based on the processed information. The Output Layer presents the recommended career paths, relevant skills, skill-gap information, and career roadmap to the user.

Figure 9.1 shows the overall architecture.

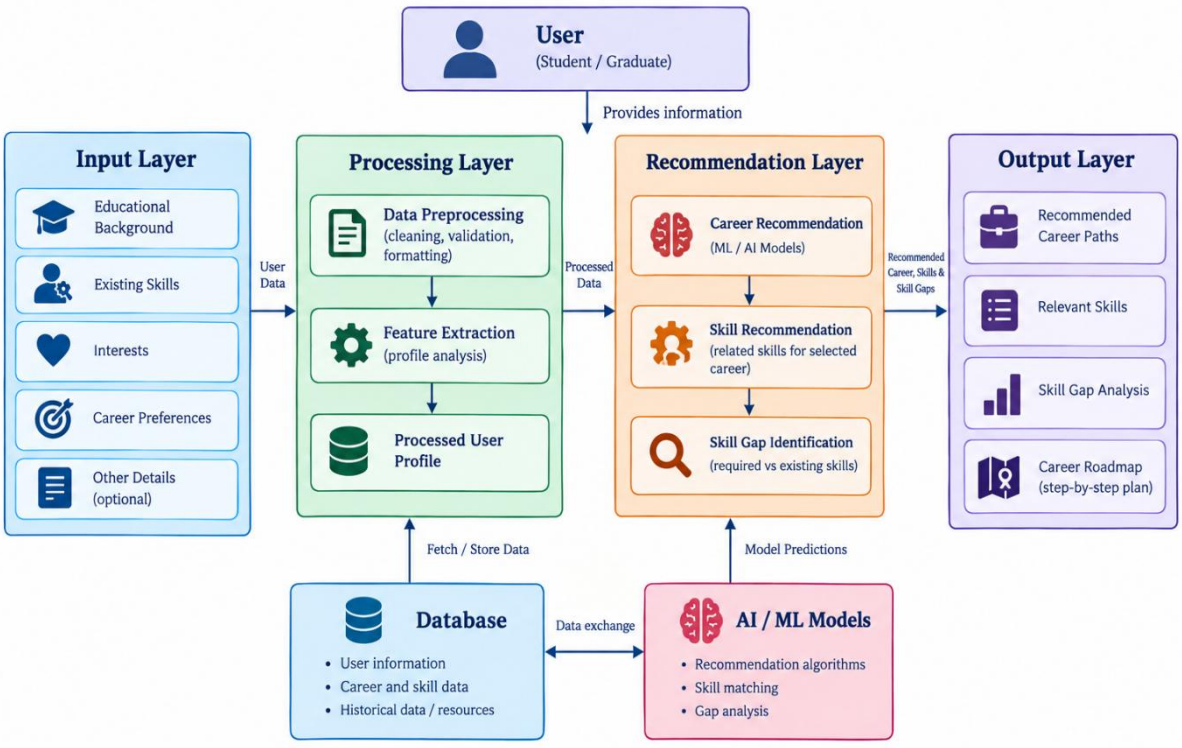


Figure: System Architecture of AI-Powered Career Roadmap and Skill Recommendation System

Figure 9.1: System architecture of AI-Powered Career Roadmap and Skill Recommendation System

The overall flow of the proposed system can be represented as follows:

- **User:** The user interacts with the system by providing the required information related to their education, skills, interests, and career preferences.
- **Input Layer:** This layer collects and organizes the information provided by the user.
- **Processing Layer:** The collected information is processed to identify relevant characteristics of the user's profile.
- **Recommendation Layer:** The processed information is used to determine suitable career options and recommend the skills relevant to those career paths.
- **Skill Gap Identification**: The system helps identify areas where the user's existing skills may need further development in relation to the recommended career.
- **Career Roadmap Generation:** Based on the recommended career and skills, the system provides a structured roadmap to guide the user's career and skill development.
- **Output Layer:** The final recommendations and roadmap are presented to the user in an understandable format.

9.2 Data Flow

The data flow of your system can be explained in this order:

User → Web Application → Data Processing → Recommendation Module → Output

The user provides information such as **educational background, existing skills, interests, and career preferences**. The web application collects and validates this information before sending it to the processing module. The processing module prepares and analyzes the user information.

The processed user profile is then passed to the recommendation module. The recommendation module analyzes the profile and generates **career recommendations, relevant skill ecommendations, and skill-gap information**. Based on these results, the system generates a **personalized career roadmap**.

The required user, career, and skill information can be stored and retrieved from the database. .Finally, the generated recommendations and roadmap are presented to the user through the application

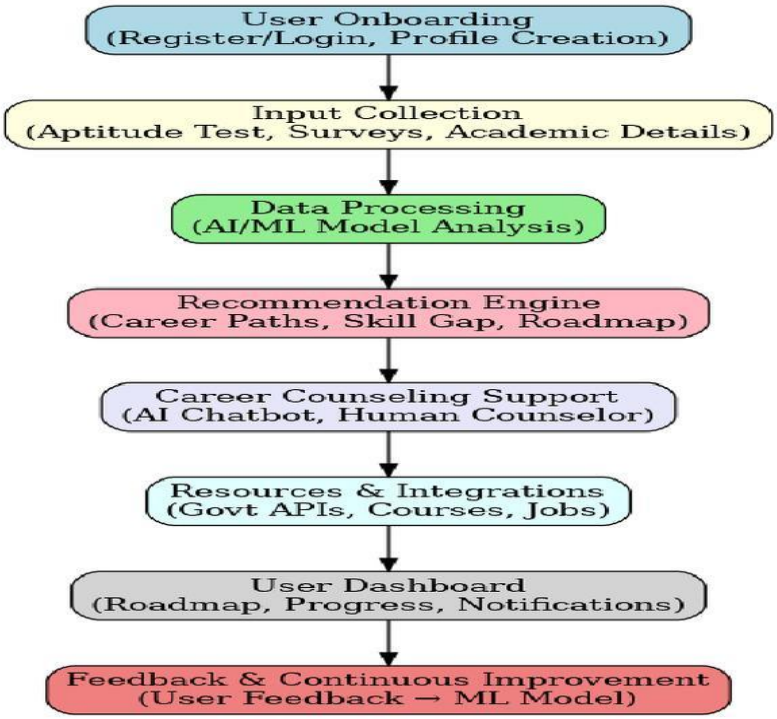


Figure 9.3: Data flow diagram of AI-Powered Career Roadmap and Skill Recommendation System

CHAPTER 10: MODULES

10.1 Module Overview

The proposed system is organized into several modules, with each module performing a specific function in the career guidance and recommendation process. The modules work together to collect user information, process the data, generate career and skill recommendations, identify skill gaps, and provide a personalized career roadmap.

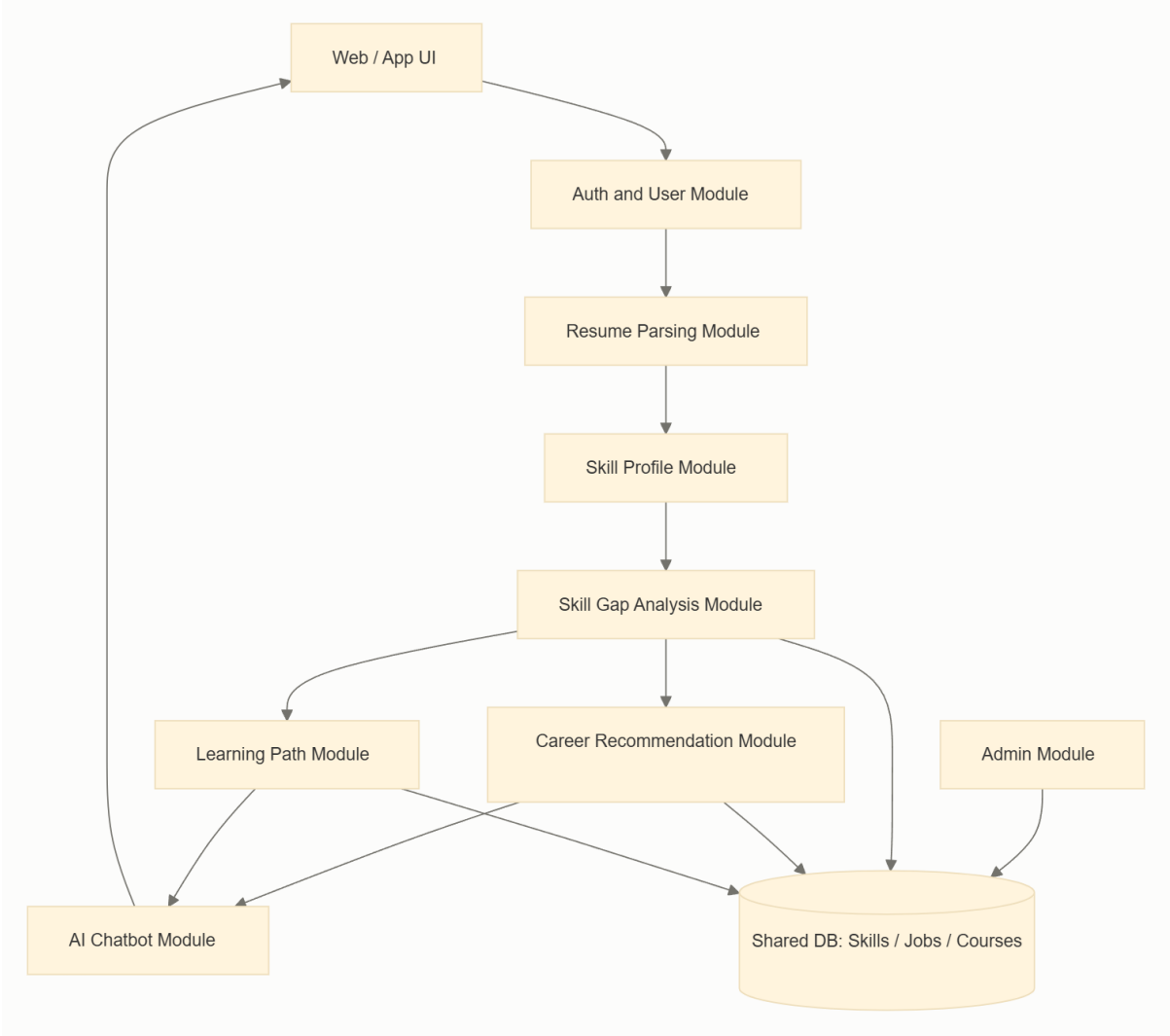


Figure 10.1: Module-level flow of AI-Powered Career Roadmap and Skill Recommendation System

10.2 Description of Modules

1. **Module 1: User Registration and Authentication Module**

   This module handles user onboarding and access control. It allows a new user to register with basic details (name, email, academic background) and enables secure login for returning users through credential-based or token-based authentication. It also maintains session management so that a user's profile and progress are preserved across visits. This module forms the entry point to the system and ensures that all subsequent modules operate on an authenticated, identifiable user.

2. **Module 2 : Resume Upload and Parsing Module**

   This module allows the user to upload their resume in common formats (PDF/DOCX) and processes it using Natural Language Processing techniques. It performs text extraction followed by Named Entity Recognition (NER) to identify structured fields such as name, education, work experience, certifications, and technical/soft skills from unstructured resume text. The output of this module is a cleaned, structured representation of the resume that feeds directly into the skill-profiling module.

3. **Module 3 : Skill Extraction and Profile Building Module**

   Using the structured data from the parsing module, this module constructs a Skill Profile for the user — a consolidated list of current skills, proficiency indicators (where available), and academic/professional background. It may also allow the user to manually add or edit skills that were not captured automatically, improving the accuracy of downstream recommendations.

4. **Module 4 : Skill Gap Analysis Module**

   This module compares the user's current Skill Profile against the skill requirements of a selected or predicted target role. It uses a predefined or continuously updated skill taxonomy (mapping job roles to required competencies) to compute the difference between "have" and "need" skills. The output is a ranked list of missing or weak skills, which becomes the basis for both career recommendations and learning-path generation.

5. **Module 5 : Career Recommendation Module**
   This is the core decision-making module. Based on the user's skill profile, interests, and academic performance, it applies a classification or similarity-matching model (e.g., Random Forest, content-based filtering, or an LLM-based reasoning layer) to recommend suitable career paths or job roles, each typically accompanied by a match score and a brief justification of why that role fits the user's profile.
6. **Module 6 : Personalized Learning Path / Roadmap Generation Module**
   Once skill gaps are identified, this module sequences relevant courses, certifications, or learning resources into a structured, step-by-step roadmap. It may use collaborative filtering, content-based filtering, or knowledge-graph-based sequencing to order resources by prerequisite dependency and difficulty level, helping the user progress logically from their current skill level to their target role.
7. **Module 7:. AI Chatbot / Conversational Counseling Module**
   This module provides a natural-language interface (often LLM-powered) that allows the user to ask follow-up questions about their recommended career path, required skills, or learning roadmap. It can dynamically generate clarifying questions, explain why a particular recommendation was made, and offer alternative suggestions, making the guidance process feel more interactive than a static report.
8. **Module 8 : Admin and Database Management Module**
   This module is restricted to administrative users and allows them to maintain the backend data that powers the entire system — updating the job-role/skill taxonomy, managing the course/learning-resource database, and monitoring overall platform usage. Keeping this as a separate module ensures the recommendation logic always operates on current, curated data.
9. **Module 9 :Dashboard and Visualization Module** ***(optional but commonly included)***
   This module presents the user's skill profile, skill-gap analysis, and roadmap progress through visual elements such as progress bars, radar/spider charts for skill comparison, and a timeline view of the learning roadmap, making the output easier to interpret than plain text.

CHAPTER 11: UML DIAGRAMS

11.1 Use Case Diagram

The use case diagram represents the functional scope of the AI-Powered Career Roadmap and Skill Recommendation System from the perspective of its two actors: the Student and the Admin. The Student, who is the primary user, interacts with the system to register or log in, upload a resume, analyze the skill gap between their current profile and a target role, receive personalized career recommendations, view a structured learning roadmap, and converse with an AI-based career counselor for further guidance. The Admin actor is responsible for maintaining the backend data that powers these recommendations, specifically managing the job-role.

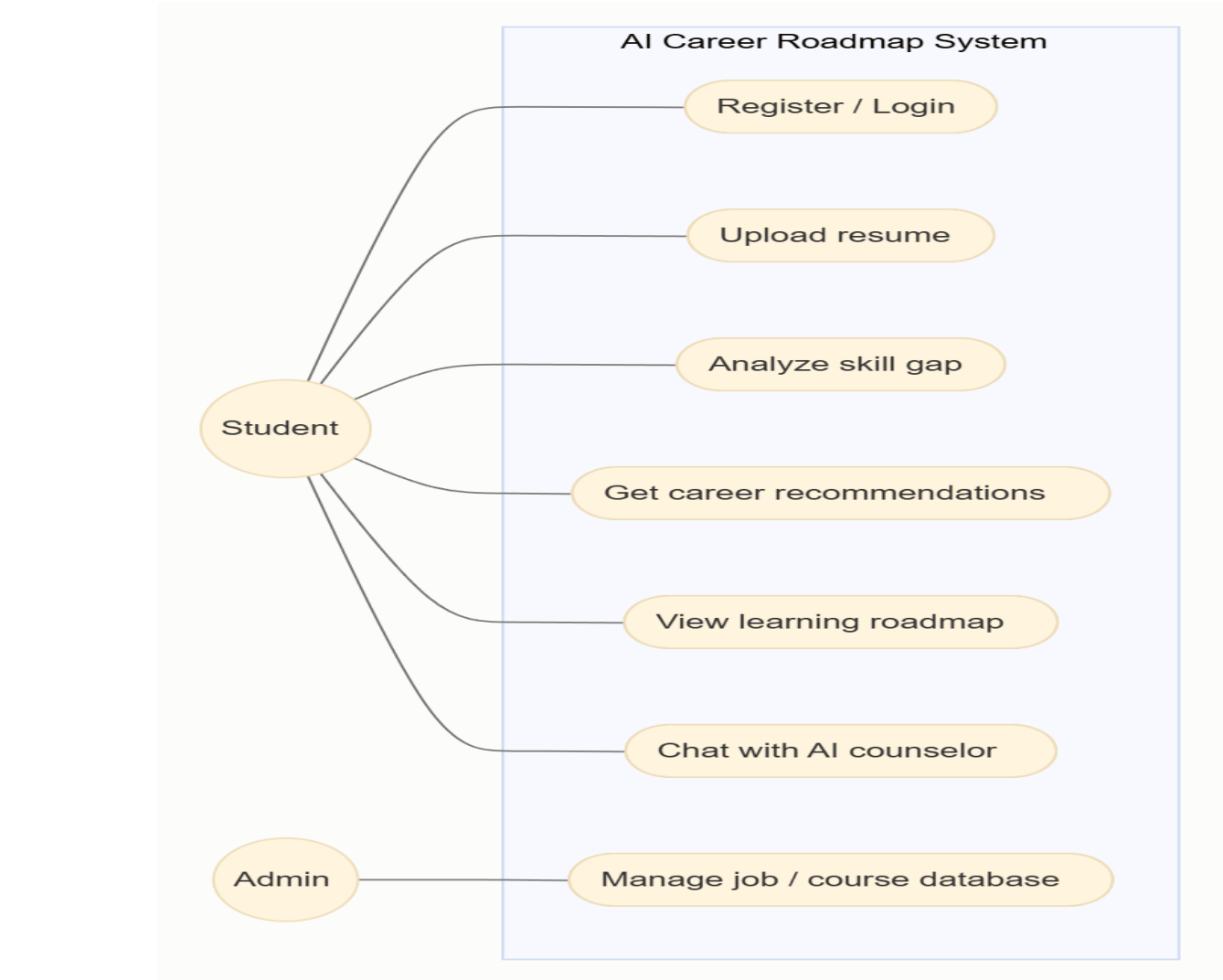


Figure 11.1: Use case diagram

11.2 Class Diagram

The class diagram models the static structure of the system by identifying its core entities, their attributes, and the relationships between them. The User class stores identity information and exposes registration and login behavior. Each User uploads a Resume, which holds the file path and a list of extracted skills obtained after parsing. The extracted skills populate a SkillProfile, which maintains both the user's current skills and the target skills required for a chosen role, and computes the gap between them..

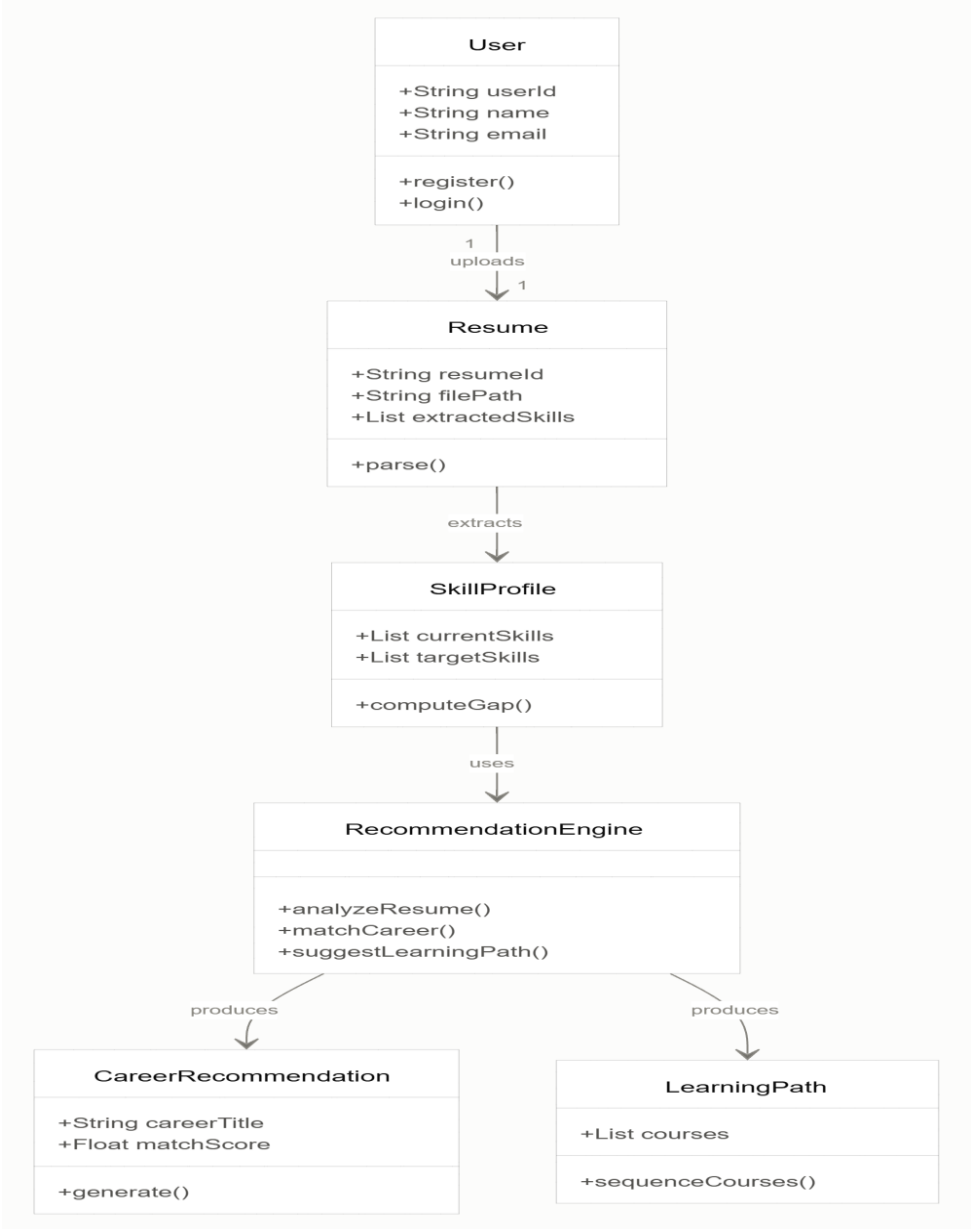


Figure 11.2: Class diagram

11.3 Sequence Diagram

The sequence diagram illustrates the time-ordered interaction that occurs when a Student requests a career roadmap. The process begins when the Student uploads a resume through the Web/App UI, which forwards the file to the NLP Resume Parser. The parser performs Named Entity Recognition internally to extract skills, education, and experience, and passes this structured data to the Recommendation Engine. The engine then queries the Skills/Jobs Database to fetch the skill requirements of the target or best-fit role, computes the skill gap internally, and returns a combined response — career suggestions along with a learning roadmap — back to the UI, which finally displays the result to the Student.

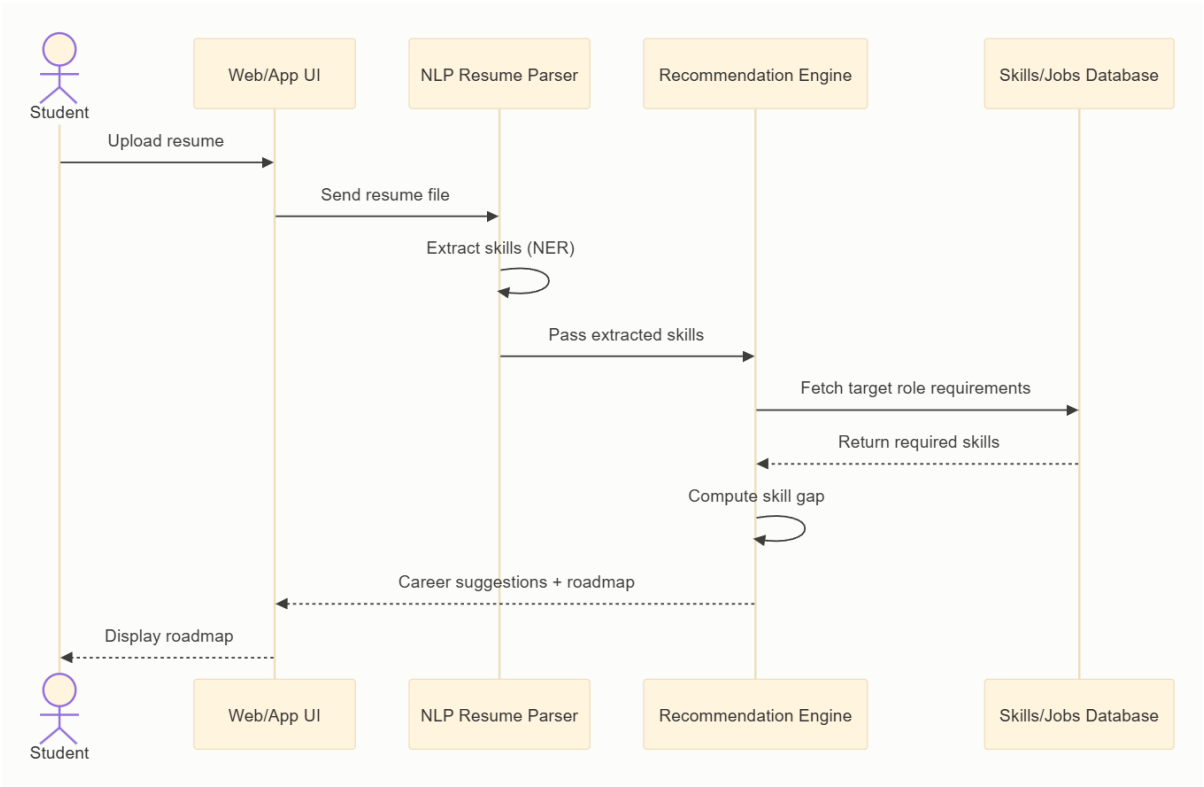


Figure 11.3: Sequence diagram for a document-grounded question

11.4 Activity Diagram

The activity diagram captures the step-by-step control flow of the system from the user's point of view, including the key decision point that differentiates the system's two possible output paths. After the Student logs in and uploads their resume, the system parses the document using NLP and extracts the relevant skills. It then compares these extracted skills against the requirements of the target role.

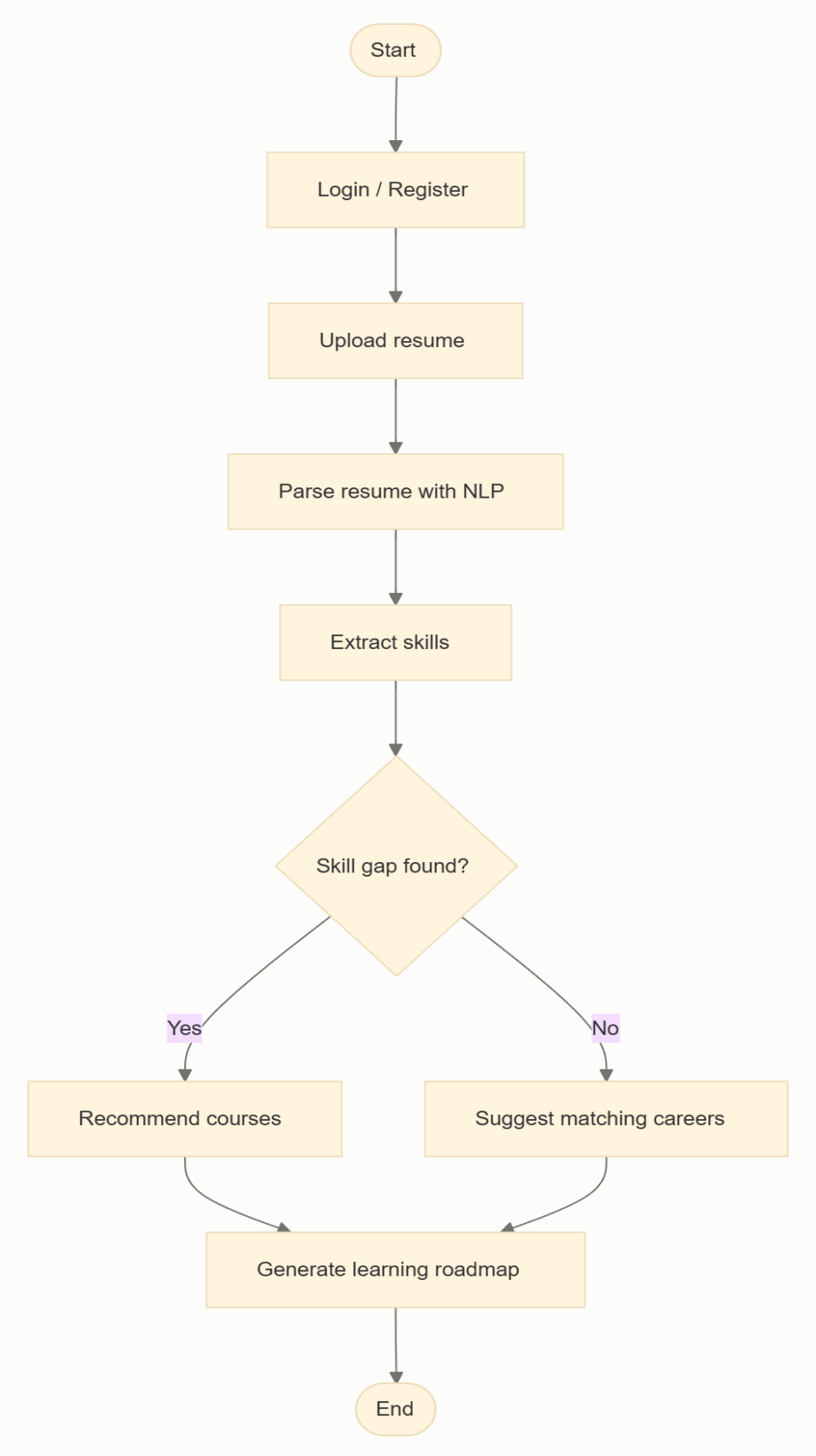


Figure 11.4: Activity diagram

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